## ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** acute | Compare $30^\circ$ with $90^\circ$ and $180^\circ$. |
| 2 | **(d)** 60 | The hour hand turns $30^\circ$ in 1 hour. |
| 3 | **(d)** 90 | A full turn is $360^\circ$. |
| 4 | **(b)** 3 | Count every pair of points. |
| 5 | **(d)** two end points | Definition |
| 6 | **(b)** 30 | In 1 minute the minute hand turns $6^\circ$. |
| 7 | **(a)** right | Compare $90^\circ$ with $90^\circ$ and $180^\circ$. |
| 8 | **(c)** 90 | The hour hand turns $30^\circ$ in 1 hour. |
| 9 | **(b)** 180 | A full turn is $360^\circ$. |
| 10 | **(c)** 6 | Count every pair of points. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | It is acute (1); Compare with the right angle $90^\circ$ (and the straight angle $180^\circ$) (1). | 1+1 |
| 12 | It turns $30^\circ$ per hour (1); $3 \times 30 = 90^\circ$ (1). | 1+1 |
| 13 | Each pair of points gives a segment (1); 7 points give 21 segments (1). | 1+1 |
| 14 | It is right (1); Compare with the right angle $90^\circ$ (and the straight angle $180^\circ$) (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | *quarter* of $360^\circ$ (1); $= 90^\circ$ (1); Type: right (1). | 1+1+1 |
| 16 | Each hour mark is $30^\circ$ apart (1); Angle $= 1 \times 30 = 30^\circ$ (1); Type: acute (1). | 1+1+1 |
| 17 | It turns $6^\circ$ per minute (1); $8 \times 6 = 48^\circ$ (1); Less than $90^\circ$ (1). | 1+1+1 |
| | **OR:** *half* of $360^\circ$ (1); $= 180^\circ$ (1); Type: straight (1). | 1+1+1 |
| 18 | Each hour mark is $30^\circ$ apart (1); Angle $= 2 \times 30 = 60^\circ$ (1); Type: acute (1). | 1+1+1 |
| | **OR:** It turns $6^\circ$ per minute (1); $12 \times 6 = 72^\circ$ (1); Less than $90^\circ$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $30^\circ$ is acute (1); $360 \div 30 = 12$ (1); 12 turns make a full turn (1); Check: product is $360^\circ$ (2). | 1+1+1+2 |
| | **OR:** Segments: $4 \times 3 \div 2 = 6$ (1); A can be paired with each of the other 3 points (1); 3 segments have A as an end point (1); A segment has two end points and a fixed length (2). | 1+1+1+2 |
| 20 | $45^\circ$ is acute (1); $360 \div 45 = 8$ (1); 8 turns make a full turn (1); Check: product is $360^\circ$ (2). | 1+1+1+2 |
| | **OR:** Segments: $5 \times 4 \div 2 = 10$ (1); A can be paired with each of the other 4 points (1); 4 segments have A as an end point (1); A segment has two end points and a fixed length (2). | 1+1+1+2 |

**Mistakes to avoid:**

- Reading the wrong scale on the protractor (inner vs outer). Decide first whether the angle is acute or obtuse.
- Using $30^\circ$ per minute for the minute hand; it is $6^\circ$ per minute, and $30^\circ$ per hour is for the hour hand.
- Calling a $200^\circ$ or $225^\circ$ angle obtuse. Obtuse angles are less than $180^\circ$; larger ones are reflex.
- Saying a ray has two end points; a ray has exactly one end point.
- Counting line segments for $n$ points as $n(n-1)$ without dividing by 2 (segment AB and BA are the same).
- Placing the protractor's centre off the vertex, or starting the scale from the wrong arm.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** one starting point | Definition |
| 2 | **(d)** 60 | In 1 minute the minute hand turns $6^\circ$. |
| 3 | **(b)** obtuse | Compare $135^\circ$ with $90^\circ$ and $180^\circ$. |
| 4 | **(d)** 120 | The hour hand turns $30^\circ$ in 1 hour. |
| 5 | **(a)** 270 | A full turn is $360^\circ$. |
| 6 | **(d)** 10 | Count every pair of points. |
| 7 | **(a)** parallel | Definition |
| 8 | **(d)** 90 | In 1 minute the minute hand turns $6^\circ$. |
| 9 | **(a)** straight | Compare $180^\circ$ with $90^\circ$ and $180^\circ$. |
| 10 | **(d)** 150 | The hour hand turns $30^\circ$ in 1 hour. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | It turns $30^\circ$ per hour (1); $5 \times 30 = 150^\circ$ (1). | 1+1 |
| 12 | Each pair of points gives a segment (1); 8 points give 28 segments (1). | 1+1 |
| 13 | It is obtuse (1); Compare with the right angle $90^\circ$ (and the straight angle $180^\circ$) (1). | 1+1 |
| 14 | It turns $30^\circ$ per hour (1); $5 \times 30 = 150^\circ$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $three - quarter$ of $360^\circ$ (1); $= 270^\circ$ (1); Type: reflex (1). | 1+1+1 |
| 16 | Each hour mark is $30^\circ$ apart (1); Angle $= 3 \times 30 = 90^\circ$ (1); Type: right (1). | 1+1+1 |
| 17 | It turns $6^\circ$ per minute (1); $18 \times 6 = 108^\circ$ (1); More than $90^\circ$ (1). | 1+1+1 |
| | **OR:** $one - third$ of $360^\circ$ (1); $= 120^\circ$ (1); Type: obtuse (1). | 1+1+1 |
| 18 | Each hour mark is $30^\circ$ apart (1); Angle $= 4 \times 30 = 120^\circ$ (1); Type: obtuse (1). | 1+1+1 |
| | **OR:** It turns $6^\circ$ per minute (1); $22 \times 6 = 132^\circ$ (1); More than $90^\circ$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $60^\circ$ is acute (1); $360 \div 60 = 6$ (1); 6 turns make a full turn (1); Check: product is $360^\circ$ (2). | 1+1+1+2 |
| | **OR:** Segments: $6 \times 5 \div 2 = 15$ (1); A can be paired with each of the other 5 points (1); 5 segments have A as an end point (1); A segment has two end points and a fixed length (2). | 1+1+1+2 |
| 20 | $72^\circ$ is acute (1); $360 \div 72 = 5$ (1); 5 turns make a full turn (1); Check: product is $360^\circ$ (2). | 1+1+1+2 |
| | **OR:** Segments: $7 \times 6 \div 2 = 21$ (1); A can be paired with each of the other 6 points (1); 6 segments have A as an end point (1); A segment has two end points and a fixed length (2). | 1+1+1+2 |

**Mistakes to avoid:**

- Reading the wrong scale on the protractor (inner vs outer). Decide first whether the angle is acute or obtuse.
- Using $30^\circ$ per minute for the minute hand; it is $6^\circ$ per minute, and $30^\circ$ per hour is for the hour hand.
- Calling a $200^\circ$ or $225^\circ$ angle obtuse. Obtuse angles are less than $180^\circ$; larger ones are reflex.
- Saying a ray has two end points; a ray has exactly one end point.
- Counting line segments for $n$ points as $n(n-1)$ without dividing by 2 (segment AB and BA are the same).
- Placing the protractor's centre off the vertex, or starting the scale from the wrong arm.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 120 | A full turn is $360°$. |
| 2 | **(b)** 15 | Count every pair of points. |
| 3 | **(c)** perpendicular | Definition |
| 4 | **(c)** 120 | In 1 minute the minute hand turns $6°$. |
| 5 | **(c)** reflex | Compare $250°$ with $90°$ and $180°$. |
| 6 | **(b)** 210 | The hour hand turns $30°$ in 1 hour. |
| 7 | **(b)** 60 | A full turn is $360°$. |
| 8 | **(c)** no end points | Definition |
| 9 | **(a)** 150 | In 1 minute the minute hand turns $6°$. |
| 10 | **(a)** acute | Compare $65°$ with $90°$ and $180°$. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Each pair of points gives a segment (1); $9$ points give $36$ segments (1). | 1+1 |
| 12 | It is reflex (1); Compare with the right angle $90°$ (and the straight angle $180°$) (1). | 1+1 |
| 13 | It turns $30°$ per hour (1); $4 \times 30 = 120°$ (1). | 1+1 |
| 14 | Each pair of points gives a segment (1); $10$ points give $45$ segments (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $one - sixth$ of $360°$ (1); $= 60°$ (1); Type: acute (1). | 1+1+1 |
| 16 | Each hour mark is $30°$ apart (1); Angle $= 5 \times 30 = 150°$ (1); Type: obtuse (1). | 1+1+1 |
| 17 | It turns $6°$ per minute (1); $35 \times 6 = 210°$ (1); More than $90°$ (1). | 1+1+1 |
| | **OR:** $one - eighth$ of $360°$ (1); $= 45°$ (1); Type: acute (1). | 1+1+1 |
| 18 | Each hour mark is $30°$ apart (1); Angle $= 6 \times 30 = 180°$ (1); Type: straight (1). | 1+1+1 |
| | **OR:** It turns $6°$ per minute (1); $40 \times 6 = 240°$ (1); More than $90°$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $90°$ is right (1); $360 \div 90 = 4$ (1); 4 turns make a full turn (1); Check: product is $360°$ (2). | 1+1+1+2 |
| | **OR:** Segments: $8 \times 7 \div 2 = 28$ (1); A can be paired with each of the other 7 points (1); 7 segments have A as an end point (1); A segment has two end points and a fixed length (2). | 1+1+1+2 |
| 20 | $120°$ is obtuse (1); $360 \div 120 = 3$ (1); 3 turns make a full turn (1); Check: product is $360°$ (2). | 1+1+1+2 |
| | **OR:** Segments: $9 \times 8 \div 2 = 36$ (1); A can be paired with each of the other 8 points (1); 8 segments have A as an end point (1); A segment has two end points and a fixed length (2). | 1+1+1+2 |

**Mistakes to avoid:**

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- Calling a $200°$ or $225°$ angle obtuse. Obtuse angles are less than $180°$; larger ones are reflex.
- Saying a ray has two end points; a ray has exactly one end point.
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